

A Versatile Foundational Language Model for Plant Cis-Regulatory Elements

Introduction

The precise regulation of the spatiotemporal expression of genes forms the foundation for life's growth, development, reproduction, and response to environmental cues. Cis-regulatory elements (CREs), non-coding sequences within the genome, precisely govern the initiation, rate, and direction of transcription by recruiting transcription factors (TFs) and synergistically interacting with other regulatory factors. Unraveling their intricate mechanisms is fundamental for understanding life processes¹⁻³ and has promising applications for crop breeding⁴⁻⁶ and the medical field^{7,8}. A deeper understanding of the complex regulatory mechanisms orchestrated by CREs, coupled with the ability to design them *de novo*, could profoundly advance synthetic biology, crop genetic improvement, and biotechnology innovation.

However, traditional biological approaches face significant challenges in identifying and deciphering CREs. The complex grammatical rules that govern CRE function, compounded by their organization into multi-CRE clusters known as cis-regulatory modules (CRMs), make large-scale experimental identification and functional characterization of CREs particularly difficult⁹.

With the rapid development of deep learning, attention has increasingly shifted towards leveraging its capabilities for predicting and designing CREs¹⁰⁻²⁴. Specifically for *de novo* design, current strategies typically involve training predictive models for regulatory element activity, followed by iterative optimization using genetic algorithms. However, this approach suffers from several limitations: it requires large volumes of high-quality training data, limiting applicability in few-shot scenarios and constraining generalization capabilities. Furthermore, even subtle inaccuracies in the predictive model may be amplified during iterative optimization, potentially leading to generated CREs that deviate significantly from the desired functional outcome.

Furthermore, recent efforts have explored the usage of language models to capture genomic information by pre-training on whole genomes and subsequently applying the models to CREs studies^{22,25-30}. While multi-species or plant-specific genomic language models, excel in contextual modeling and semantic understanding due to their masked language model (MLM) architectures³¹, they inherently lack *de novo* generative capabilities³². Crucially, since CREs constitute only a small fraction of the genome, training MLMs on entire genome with limited parameters may result in suboptimal learning of the intricate grammar and fine-grained patterns unique to CREs.

To overcome these limitations—namely the reliance on predictive model—genetic

algorithm pipelines with limited generalization, and the lack of direct generative capacity in genome models based on masked language model (MLM) architectures. we developed VERDVersatile Engine for Regulatory Design, a novel framework for plant CREs research and design, built upon an autoregressive Transformer-decoder architecture and serving as a foundational language model for these elements. VERD is trained on a meticulously curated, high-quality CRE dataset derived from chromatin accessibility data across diverse plant species and tissues, with the explicit aim of capturing the intrinsic grammatical features of plant regulatory elements. Unlike MLMs, VERD's autoregressive design confers robust generative capabilities enabling the direct de novo design of functional regulatory element sequences, requiring only minimal fine-tuning on small datasets. This approach obviates the need for external iterative optimization algorithms, thereby reducing error accumulation and enhancing generalization. VERD matches or exceeds existing MLM-based models in plant CRE functional characterization, while achieving comparable accuracy to specialized predictors using just 1/500 of the training data. We also developed interpretability tools revealing strong alignment between VERD's learned regulatory grammar and known biology. Notably, VERD enables efficient de novo design of plant regulatory elements with minimal fine-tuning, achieving a 90% success rate in core promoter design upon experimental validation.

VERD is a Foundational language model of CREs in plants

To model the cis-regulatory elements (CREs) in plants, we constructed a CRE dataset spanning phylogenetically diverse species. Unlike other genomic elements such as genes, CREs are poorly defined in genomes due to their lack of sequence features. To accurately define CREs, we leveraged DNase-seq and ATAC-seq data to identify accessible chromatin regions in each species, which are considered as a mark of functional CREs^{33,34}. The resulting CRE dataset comprises CREs from 19 tissues, covering 20 dicots and 9 monocots (Figure 1A and supplementary Table X). This CRE dataset provides a comprehensive representation of regulatory landscapes in plant species.

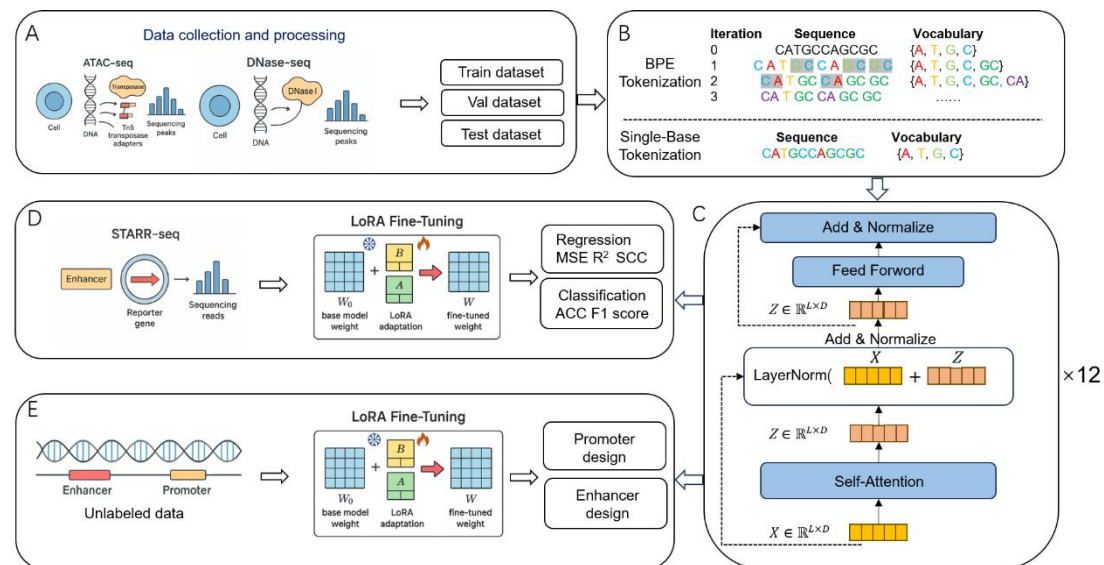


Figure 1:

CREs contain clusters of transcription factor (TF) binding sites that function cooperatively to produce diverse regulatory outcomes⁹. TVERD To systematically decode these combinatorial rules, we trained VERD using a Transformer-decoder architecture³⁵. VERDLeveraging multi-head attention, the model captures both local and global grammatical patterns within CREs, enabling effective learning of the complex combinatorial grammars of TF binding sites. Through the integration of an autoregressive strategy, it can also design novel sequencesVERD based on learned representations. VERDTrained on the 2,696,144 CREs of 2.2 billion DNA base pairs using single-base and Byte Pair Encoding (BPE) tokenization for various downstream tasks²⁷ (See Supplementary notes). VERD supports diverse downstream tasks, including the prediction of CRE activity, the generation of CREs, and the interpretation of mutations within CREs.

Feature extraction with VERD captures regulatory grammar and identifies novel accessible chromatin regions

To evaluate the model's capacity to learn regulatory element sequences, we first assembled a dataset comprising regulatory element sequences from eight diverse plant species. Both VERD-single and VERD-BPE were then appliedVERDVERD for feature extraction and embedding generation. T-SNE analysis of the resulting embedding features revealed that both VERD-single and VERD-BPE effectively clustered regulatory element sequences from the same species into distinct regions (Figure 2A, B). Furthermore, monocot and dicot regulatory elements were well separated into two clusters. This striking separation demonstrates VERD's successful learning of fundamental features of diverse plant regulatory elements, and its implicit capture of their evolutionary relationships. Concurrently, significant overlaps across species suggest considerable conservation within plant regulatory landscapes. This observation further supports the rationale behind employing pre-trained models to learn general principles of plant regulatory elements, followed by fine-tuning on specific, smaller datasets.

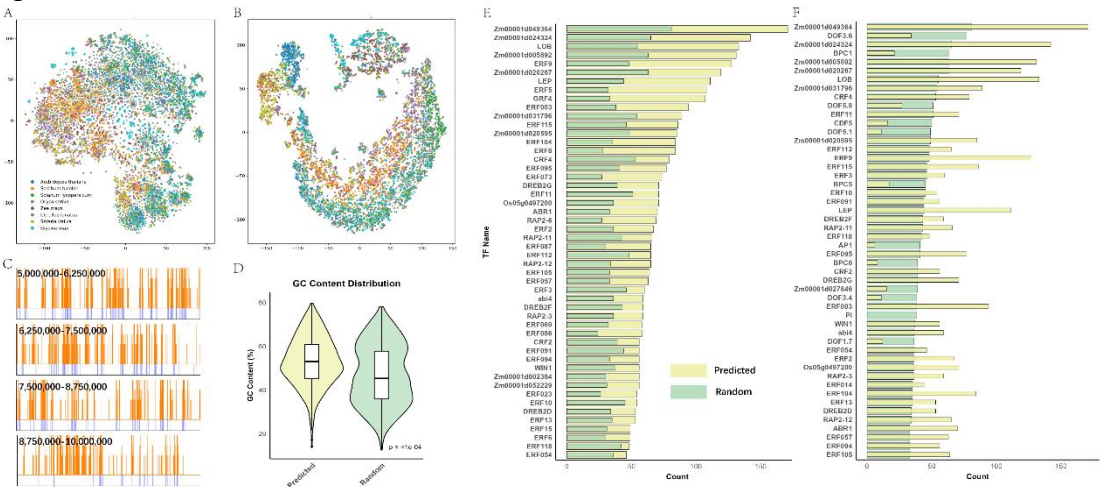


Figure 2

To ascertain whether the model acquired intrinsic features of different regulatory element types during pre-training, an additional dataset was constructed comprising three distinct maize regulatory element types: promoters, enhancers, and distal regulatory elements (Supplemental Figure). T-SNE analysis revealed the model's approximate ability to distinguish features of these different regulatory element types. Specifically, VERD-single demonstrated higher sensitivity to distal regulatory sequences, whereas VERD-BPE exhibited a degree of discriminatory capability across all three regulatory element types.

Furthermore, we investigated VERD's capacity to predict genome-wide regulatory element information. In this regard, we fine-tuned the model using chromatin accessibility data from chromosome 1 of *Zea mays* (Mo17 accession) and subsequently predicted accessible region distributions across other chromosomes. As the Mo17 genome was notably not part of the pre-training dataset, this assessment also served to rigorously evaluate VERD's generalization capabilities on unseen data. The results indicated that VERD's predictions of chromatin accessible regions not only largely encompassed experimentally validated regions but also successfully identified novel regulatory element areas previously undetected through experimental means (Figure 2C).

To further delve into the nature of these newly identified regions, we investigated whether VERD's predicted chromatin accessible regions possessed distinct internal characteristics compared to non-accessible sequences. Specifically, we observed that numerous segments within the 9M-9.25M region of Mo17 chromosome 2 were predicted as chromatin accessible by VERD, a region where experimental assays had not previously detected accessibility signals. A comparison of these newly predicted accessible segments with random sequences from the same genomic locus revealed that the accessible sequences exhibited significantly higher GC content than the random sequences (Figure 2D), a finding consistent with established genomic principles.³⁶ Moreover, these fragments were enriched for transcription factor binding motifs (Figure 2E,F). This enrichment suggests that these regions are more likely to be involved in gene expression regulation through the recruitment and binding of transcription factors.

VERD-Single was further applied to predict the effects of single nucleotide polymorphisms (SNPs) within *Zea mays* Mo17 chromatin accessible regions. Logit differences at SNP loci (Supplemental Figure) were calculated to estimate the impact of them on the structural and functional stability of cis-regulatory elements (CREs). Specifically, a larger logit difference indicates a more substantial effect of a given SNP on CRE structure and function. This approach offers a novel avenue for investigating and leveraging SNPs located within regulatory regions.

VERD achieves high-accuracy prediction across diverse regulatory tasks

We first evaluated VERD's ability to predict the strength of promoters, a crucial factor

in controlling transcription initiation. For this, we utilized a recently published dataset, which quantified promoter strength across three species in maize and tobacco (see Methods). VERD successfully predicted promoter strength, achieving R^2 values of 0.736 and 0.706 for tobacco leaf and maize protoplast, respectively. The Spearman correlation coefficients, reflecting the concordance between predicted and observed values, were 0.858 for leaf and 0.821 for protoplast. These results matched or even exceeded the performance of current state-of-the-art plant genome models, such as AgroNT and PDLLEs. Within the dataset, some promoters exhibited substantial variations in strength between tobacco and maize, whereas others showed similar activity. The dataset also included promoter strength measurements under different environmental conditions, specifically dark and light. Consequently, we employed VERD and two other plant genome models for classification tasks involving these specific and condition-responsive promoters. In these tasks, all three models performed well on species data, however, only VERD achieved accuracy and F1 scores near 0.95 on condition-induced data.

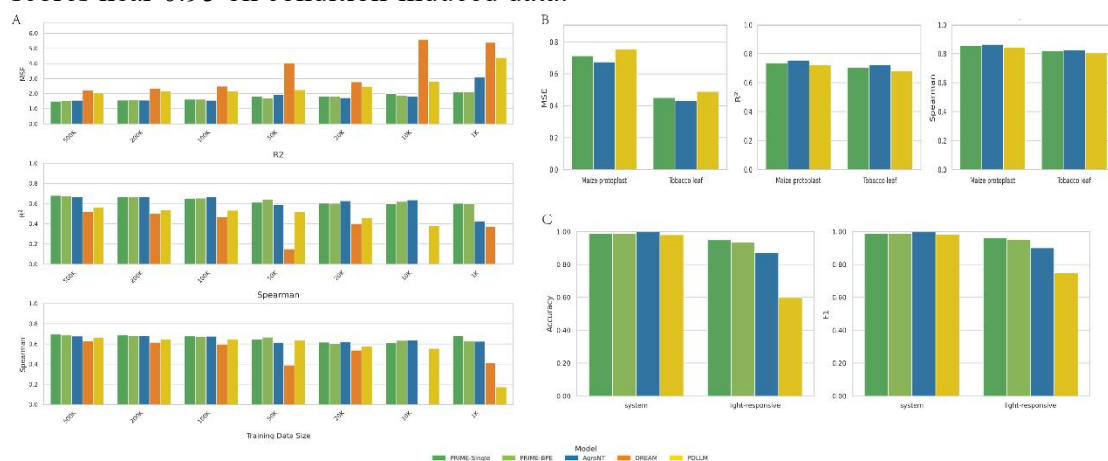


Figure 3:

Distal regulatory elements such as enhancers played essential roles in gene expression. To evaluate VERD's ability to predict enhancer strength and its few-shot learning capability, we fine-tuned VERD on datasets of varying sizes: 500,000, 200,000, 100,000, 50,000, 20,000, 10,000, and 1,000 samples. For comparison, two plant genomic language models, AgroNT and Plant NT single base, and two CNN-based models DREAM and DeepSTARR were included. VERD-single and VERD-BPE achieved comparable or superior predictive performance to Plant NT single base and DREAM, even when fine-tuned on the smallest dataset of 1,000 samples. Notably, for the 1,000-sample dataset, only VERD-single and VERD-BPE maintained an MSE below 3. By contrast, DeepSTARR consistently underperformed across all datasets, likely due to its smaller size. Overall, VERD exhibited robust predictive power for enhancer strength, its performance on large datasets was comparable to AgroNT.VERD, while retaining stable performance under few-shot conditions.

Perturbation analysis reveals VERD's learned regulatory mechanism While deep learning-based language models exhibit strong predictive performance, they often operate as “black boxes”³⁷, obscuring the biological information the model has

learned to facilitate its predictions. To improve interpretability, a perturbation-based framework was developed (see Methods). Specifically, we aimed to identify which positions and motifs within CREs significantly influence its strength. After fine-tuning of the model on labeled datasets, the framework was applied to a test set. For each sequence in this set, we systematically replaced sequential k-mer nucleotide stretches at original positions with random sequences (Figure 4A). Difference in predicted values before and after replacement then quantified to assess the importance of that position and motif for promoter strength. Analysis clearly demonstrated that positions closer to the transcription start site (TSS) exert a greater influence on promoter strength (Figure 4B, C). There are also distinct patterns across tobacco leaf and maize protoplast systems. In tobacco leaves, positions distal to the TSS exhibited lower importance scores (Figure 4B), while in maize protoplasts, importance scores gradually rose with increasing distance from the TSS, particularly beyond 80 bp upstream (Figure 4C). Our analysis also extended to the impact of perturbation-induced motif disruption on importance scores. In both tobacco leaf and maize protoplast promoter expression systems, we identified significant TA-rich and GC-rich motifs, consistent with the established roles of the TATA-box and GC-box as crucial transcriptional activation elements (Figure 4G).

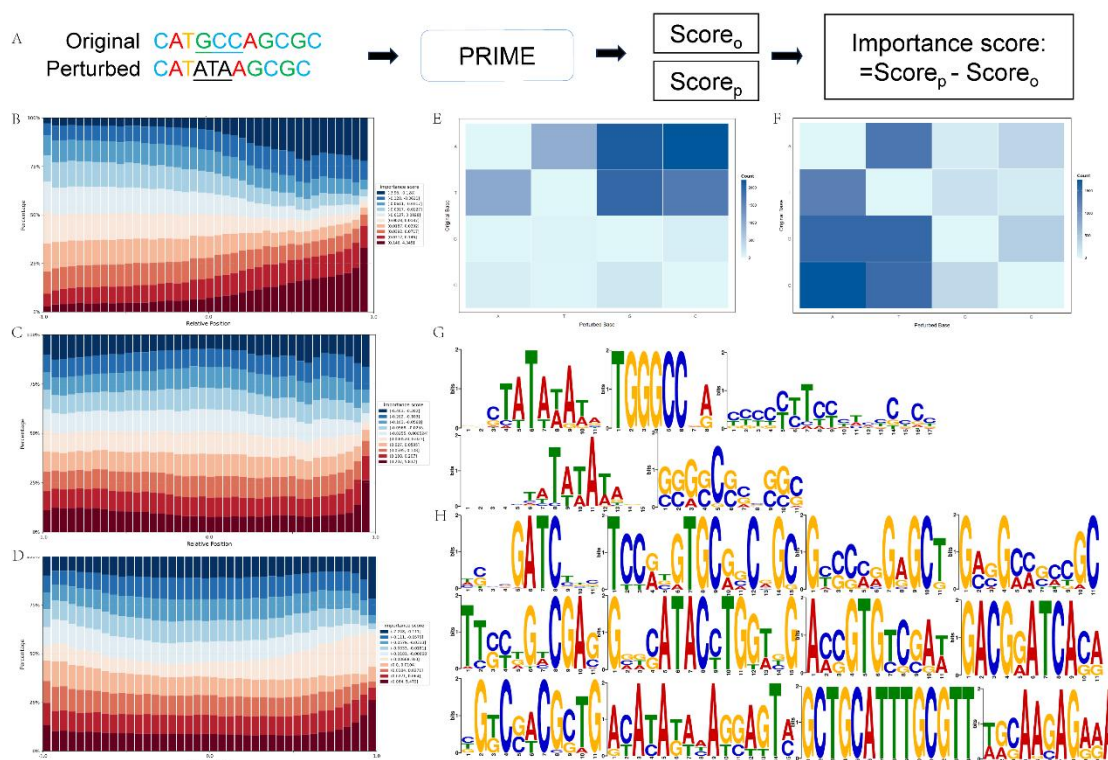


Figure 4

To further explore VERD's ability to discern subtle, yet significant, biological features, we investigated its capacity to detect the known characteristic that promoters highly expressed in tobacco leaves typically exhibit lower GC content³⁸ (Supplemental Figure). VERD To this end, we performed single-base perturbation analysis using the model fine-tuned for tobacco leaf promoter prediction. Mutations from G/C to A/T tended to increase predicted promoter strength, while A/T to G/C substitutions led to

decreased expression (Figure 4E, F), indicating that VERD has captured this characteristic of tobacco leaf promoters.

Extension of the perturbation framework to enhancers revealed that the importance of different positions for enhancer strength was relatively uniform, with a slightly higher significance observed at the distal ends compared to the central regions (Figure 4D, H). This observation aligns with previous studies indicating that transcription factor binding sites (TFBS) near the edges of enhancers may be more accessible or better positioned for interaction with co-factors and the transcription machinery (ref). Numerous motifs enriched within enhancers (Figure 4H), showed strong similarity to known transcription factor binding motifs.

De novo Design of Plant Regulatory Elements with VERD

A significant advantage of VERD, which employs an autoregressive architecture, lies in its ability to de novo design novel plant regulatory element sequences without reliance on additional genetic algorithms or similar processes. By capturing the intrinsic features of plant regulatory elements, VERD enables direct generation of sequences aligned with the desired distribution, in contrast to genetic algorithms, which rely on iterative selection from an initial seed to approximate the target sequence space (Figure 5A, B). VERD-Single was fine-tuned using 500 high-expression maize promoter sequences to generate entirely new maize promoter sequences. And we also fine-tuned regLM, another generative DNA model, with the same data. To assess the similarity between designed and original sequences, we compared the 6-mer frequency correlation and GC content between the generated sequences and the training set sequences. Additionally, we quantified the information entropy of the designed sequences to evaluate their diversity. Our design results indicated that VERD outperformed regLM in both GC content and 6-mer frequency correlation (Figure 5C, E). Furthermore, VERD also demonstrated superior diversity in its designed sequences compared to regLM, with its diversity level being comparable to that of sequences within the original training set (Figure 5F).

The same strategy was applied to maize enhancer design using VERD-BPE. Across all evaluation metrics, the results indicated that VERD-designed sequences consistently exhibited the closest resemblance to the training dataset across all evaluated metrics (Figure D-F). Notably, we also employed DREAM (trained with 500K dataset) in conjunction with genetic algorithms for sequence design. However, after 100 iterations, this approach yielded only a single repetitive sequence. This observation aligns with our hypothesis that such methods are highly dependent on the accuracy of the predictive model.

To experimentally validate the reliability of our designed sequences, we conducted promoter activity assays on 10 selected promoter sequences. Results indicated that 9 out of the 10 designed sequences exhibited promoter activity, achieving a success rate of 90%. BLAST analysis revealed no detectable homologous sequences, suggesting

that VERD genuinely de novo designed CREs with novel characteristics, rather than merely memorizing or reproducing sequences present in the training data. Notably, this high success was achieved on only 500 sequences, highlighting its strong few-shot design capability and impressive plasticity.

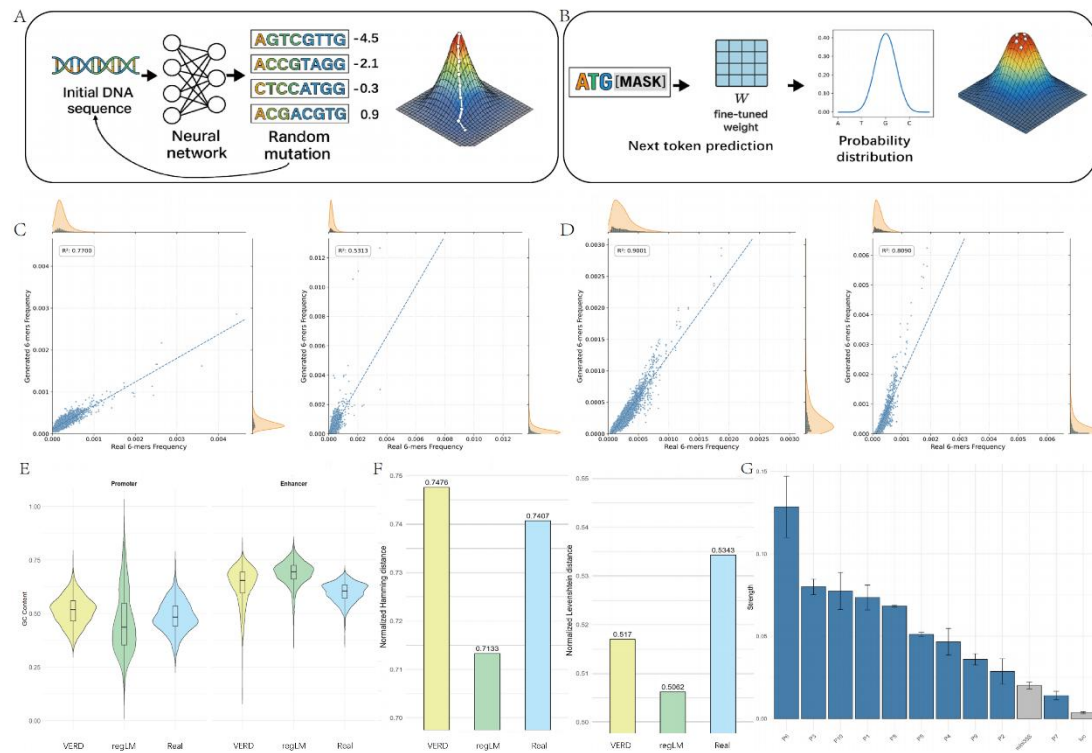


Figure 5:

Discussion

Understanding and engineering the regulatory grammar encoded in CREs is essential for advancing plant genomics and synthetic biology. In this work, we introduce VERD, an autoregressive Transformer-based foundational language model tailored specifically for plant CREs. Unlike existing genome-scale masked language models, VERD brings together strong predictive accuracy and direct generative capability, enabling both the interpretation and creation of functional regulatory sequences in a single framework.

VERD demonstrates robust performance across diverse tasks, including promoter and enhancer strength prediction, chromatin accessibility inference, and SNP impact analysis. Remarkably, it maintains high accuracy even when fine-tuned on minimal data, performing comparably to specialized predictors trained on much larger datasets. Its ability to generalize to unseen genomes highlights that VERD has learned fundamental regulatory principles rather than relying on species-specific patterns alone.

A key advantage of VERD is its built-in capacity for de novo sequence generation, eliminating the need for error-prone iterative optimization methods such as genetic

algorithms. With only modest fine-tuning, VERD generated functional promoter sequences with a 90% experimental success rate. This shows that the model not only captures regulatory rules but can also translate them into new, functional sequences with impressive efficiency and diversity.

To make these predictions biologically interpretable, we developed a perturbation-based analysis framework that pinpoints the sequence positions and motifs most critical to CRE function. VERD accurately recovered well-known regulatory signatures, including TATA and GC boxes and characteristic positional biases near transcription start sites. These results confirm that the model's internal representations align closely with established biological knowledge, making it a valuable tool for both prediction and discovery.

VERD offers a flexible foundation that can be extended to context-specific regulatory design, including applications targeting particular tissues, cell types, or environmental conditions. Such extensions would enable more precise control over gene expression and open new possibilities for crop improvement and synthetic biology.

In sum, VERD offers a unified, data-efficient, and interpretable platform that combines prediction and generation of plant regulatory elements. By bridging these two capabilities, it lays the groundwork for a new generation of computational tools to understand and design regulatory sequences in plants.

Methods

Model

In this study, we employed an autoregressive language model based on the Transformer architecture as our foundational model for learning plant regulatory element information. Autoregressive models have demonstrated immense capabilities in natural language processing due to their powerful generative capacity³⁹. Unlike masked MLMs, which predict randomly masked tokens based on surrounding context, autoregressive models predict the next token in a sequence given all preceding tokens. This sequential prediction mechanism enables the model to generate complete sequences when guided by a prompt. Both MLMs and autoregressive models leverage the multi-head attention mechanism inherent in the Transformer architecture, allowing them to capture both local and global dependencies within a given sequence.

Architecture

Our model is specifically structured as a Transformer decoder-only architecture³⁵. The tokenizer employs two distinct vocabulary types: 1) Single-base encoding: This vocabulary consists of four canonical DNA bases (A, T, C, G) and an additional 'N' for ambiguous bases. 2) Byte Pair Encoding (BPE)⁴⁰: We set a maximum vocabulary size of 4096 for this encoding. Both encoding schemes incorporate four special tokens: Begin-of-Sequence (BOS), End-of-Sequence (EOS), Padding (PAD), and Unknown (UNK). BOS marks the start of a sequence, EOS signifies its end, PAD is used to

uniformly extend sequences to the maximum input length, and UNK represents tokens not found in the vocabulary, though it is typically rarely used in our context.

The model accepts input sequences of 1024 tokens for the single-base tokenizer and 512 tokens for the BPE tokenizer. These inputs are first passed through an embedding layer, where each token is projected into a 768-dimensional space. A learned positional embedding is then added to these token embeddings to encode positional information. The resulting embedding vector is subsequently fed into 12 successive blocks. Each block comprises a multi-head attention layer, a layer normalization layer, and a multi-layer perceptron which also known as a feed-forward layer. Residual connections are strategically employed within both the attention and feed-forward layers to facilitate gradient flow and improve training stability.

Pre-training dataset

Our pre-training dataset was primarily constructed from the PlantCADB database⁴¹. This dataset encompasses ATAC-seq and DNase-seq data from 30 diverse plant species, including 20 dicots and 10 monocots, covering a range of model plants, staple crops, horticultural species, and forest trees. Raw data was first merged using bedtools v2.27.1 and mapped to their respective reference genomes, ensuring consistency with the versions specified in PlantCADB. Each contiguous accessible chromatin region was then segmented into sequences no longer than 1024 base pairs.

Pre-training model

Model pre-training was conducted on a computational node equipped with eight RTX 4090D GPUs. A data-parallel training strategy was implemented, meaning each GPU held a complete copy of the model. After processing each batch, gradients were synchronized across all GPUs, and model parameters were updated uniformly. For the single-base model, the batch size per GPU was set to 4, resulting in a total effective batch size of 32. For the BPE model, the per-GPU batch size was 8, leading to a total effective batch size of 64. Model parameters were optimized using the Adam optimizer⁴², with β_1 set to 0.9, β_2 to 0.999, and ε to 1e-8. The initial learning rate was set at 3e-5.

Fine-tune model

Upon completion of pre-training, the model was fine-tuned on various labeled datasets for specific downstream tasks. We employed the Low-Rank Adaptation (LoRA) strategy⁴³ for fine-tuning. LoRA significantly reduces the number of trainable parameters by injecting small, trainable low-rank matrices into the Transformer layers, thereby allowing for efficient adaptation to new tasks with minimal computational overhead and memory footprint, especially beneficial for large pre-trained models. Fine-tuning was performed for classification, regression, and generative tasks.

T-sne analysis for species and CREs

From the species dataset, 300 sequences were randomly sampled for each species.

Likewise, 500 sequences were randomly sampled for each CRE type in the CREs dataset. For every individual sequence, the final layer outputs of VERD-single and VERD-BPE were extracted and utilized as its embedding. t-SNE was then employed for the analysis and visualization of these embeddings.

Chromatin accessibility prediction

For the chromatin accessibility prediction task, we fine-tuned the pre-trained VERD model using data from chromosome 1 of *Zea mays* (Mo17 accession) as the training set. Experimentally validated chromatin accessible regions were labeled as positive samples (label 1), while non-accessible regions were segmented into fragments of no more than 1024 bp and labeled as negative samples. The fine-tuned model was then used to predict accessible region distributions across chromosomes 2. Specifically, we predicted a 5-10 Mbp segment of chromosome 2, omitting the initial 5 Mbp due to typically sparse accessibility in telomeric regions. Given the variable lengths of plant chromatin accessible regions, we set the model's prediction window to 256 bp, with a sliding stride of 64 bp. Consequently, each 64 bp segment, excluding the initial and final 192 bp, received four independent predictions. A segment was classified as an accessible region if it received two or more positive predictions.

In the 9–9.25 Mbp region of chromosome 2, VERD predicted multiple chromatin accessible regions that were not identified in existing experimental datasets. To investigate this discrepancy, we randomly selected an equal number of 64-bp fragments from the same region that did not overlap with the predicted accessible regions. We then compared the GC content between the predicted accessible regions and the random fragments. In addition, we used the FIMO tool⁴⁴ to scan for transcription factor (TF) binding motifs, with the parameters set as: `--verbosity 1 --bgfile --norc --thresh 1.0E-4`, and using the JASPAR2022_CORE_plants_non-redundant_v2 motif database. We further compared the abundance of the top 50 most enriched motifs between the predicted accessible region sequences and the random sequences.

SNP Effect Prediction:

For SNP effect prediction within regulatory regions, we generated a local probe sequence of 129 bp by extending 64 bp upstream and downstream from each SNP. For each SNP, two such probe sequences were created: one incorporating the reference (REF) allele and another with the alternative (ALT) allele at the SNP locus. These 129 bp sequences were then processed by VERD-Single, utilizing its pre-trained weights to obtain a logit value for each. The SNP effect was quantified as the absolute difference between the logit value of the REF allele sequence and the logit value of the ALT allele sequence, calculated as :

$$|\logit(REF) - \logit(ALT)|$$

Promoter strength prediction

For the promoter strength prediction task, we utilized a published dataset³⁸ that

quantified promoter activity in *Arabidopsis thaliana*, *Zea mays*, and *Sorghum bicolor*. Promoter strengths were validated in both tobacco leaves and maize protoplasts. Additionally, light-responsive promoters were identified within the tobacco leaf experiments. To ensure a direct comparison with existing state-of-the-art models such as AgroNT and PDLMLs, which were also evaluated on this dataset, we adopted an identical training/test set split for fine-tuning. Model performance was evaluated using the Mean Squared Error (MSE), the coefficient of determination (R^2) and the Spearman correlation coefficient.

Promoter specificity prediction

The same dataset used for promoter strength prediction was employed for evaluating promoter specificity. Promoters exhibiting an expression strength difference greater than 8-fold (i.e., a difference greater than 3 after log2 transformation) between conditions were classified as specific promoters. Conversely, for non-specific promoters, we included only those with a log2-transformed strength difference less than 0.5. For the light-responsive promoter prediction task, the dataset comprised 284 promoters highly expressed under dark conditions and 500 light-insensitive promoters. For the tissue-specific promoter task, the dataset included 400 tobacco leaf-specific promoters and 400 maize protoplast-specific promoters. Model performance was assessed using the accuracy and f1 value.

Enhancer strength prediction

For the enhancer prediction task, we utilized a recently published dataset of maize enhancers identified through STARR-seq experiments³⁶. To comprehensively evaluate model performance across varying data scales, we constructed subsets containing 500,000, 200,000, 100,000, 50,000, 20,000, 10,000, and 1,000 samples by random subsampling. MSE, R^2 and Spearman correlation were chosen as the evaluation metric for comparing different models across these datasets.

Explainability analysis

We employed a perturbation-based approach to perform model interpretability analysis. For each input sequence, starting from the first position, we randomly replaced consecutive k-mers (we set $k=3$) and slid the perturbation window across the sequence with a stride of 1. For each perturbed sequence, we predicted the original sequence score (Score_o) and the perturbed sequence score (Score_p), and calculated the importance score as:

$$\text{Importance score} = \text{Score}_p - \text{Score}_o$$

For positional importance analysis, the resulting importance scores were divided into 10 equally sized bins according to their magnitude, and for each relative position along the sequence, we calculated the proportion of scores falling into each bin. In addition, we selected samples in which perturbation resulted in more than a fourfold change in predicted score. For each such sample, we extended the k-mer perturbation region by 7 bp on both sides, yielding a $14+k$ bp region. This region was scanned for enriched motifs using the MEME tool⁴⁵, with an E-value threshold of < 0.05 , and

similar motifs were identified using Tomtom⁴⁶ against the JASPAR2022_CORE_plants_non-redundant_v2 motif database. Furthermore, we set $k=1$ to assess the impact of single-nucleotide substitutions on promoter strength in the tobacco leaf system, and performed analyses on the top 1% highest and lowest importance score perturbation samples.

Promoter and enhancer de novo design

For the promoter design task, we fine-tuned the VERD-single model. We selected 500 high-expression promoter sequences from maize protoplasts as the fine-tuning training set. A consistent prompt of "AG" was used for generation, with a target sequence length of 170 bp. After sequence generation, the initial "AG" prompt was removed, resulting in designed promoter sequences of 168 bp.

For the enhancer generation task, we fine-tuned the VERD-BPE model. We selected sequences from the top 0.1% of enhancer strengths within the dataset as our fine-tuning training set. To augment the dataset, reverse complementary sequences were also included, resulting in a final dataset of 2452 sequences. Designed sequences had a target length of 50-70 BPE tokens. A consistent prompt of "ATGC" was used for generation and subsequently removed from the final designed sequences.

We evaluated the similarity between the designed sequences and the real sequences by calculating the correlation of k-mer (we set $k=6$) frequency and GC content between the designed sequences and the training data. We computed normalized Hamming distance for designed promoters and normalized Levenshtein distance for designed enhancers to evaluate the diversity of designed sequences. The formula for calculating the normalized Hamming distance is as follows:

$$AvgND_{Hamming} = \frac{1}{\binom{N}{2}} \sum_{1 \leq i < j \leq N} \frac{D_{Hamming}(S_i, S_j)}{len(S_i)}$$

And The formula for normalized Levenshtein distance is:

$$AvgND_{Levenshtein} = \frac{1}{\binom{N}{2}} \sum_{1 \leq i < j \leq N} \frac{D_{Levenshtein}(S_i, S_j)}{\max(len(S_i), len(S_j))}$$

$D_{Hamming}(S_i, S_j)$ counts mismatches for same-length sequences, and

$D_{Levenshtein}(S_i, S_j)$ counts the minimum total insertions, deletions, and substitutions needed to transform one sequence into another, regardless of their initial lengths.

References:

1. Guo, J., Liu, M., Liu, F. & Wang, L. Jund orchestrates cis-regulatory element dynamics to facilitate endothelial-to-hematopoietic transition. *Proceedings of the*

National Academy of Sciences 122, e2426714122 (2025).

2. Sarropoulos, I. *et al.* Developmental and evolutionary dynamics of cis-regulatory elements in mouse cerebellar cells. *Science* 373, eabg4696 (2021).
3. Patrick, R. *et al.* The activity of early-life gene regulatory elements is hijacked in aging through pervasive AP-1-linked chromatin opening. *Cell Metabolism* 36, 1858-1881.e23 (2024).
4. Song, X. *et al.* Targeting a gene regulatory element enhances rice grain yield by decoupling panicle number and size. *Nat Biotechnol* 40, 1403–1411 (2022).
5. Chen, W. C. W. *et al.* A synthetic transcription platform for programmable gene expression in mammalian cells. *Nat Commun* 13, 6167 (2022).
6. Yasmeen, E., Wang, J., Riaz, M., Zhang, L. & Zuo, K. Designing artificial synthetic promoters for accurate, smart, and versatile gene expression in plants. *Plant Communications* 4, 100558 (2023).
7. Thomas, E. D. *et al.* Cell-specific cis-regulatory elements and mechanisms of non-coding genetic disease in human retina and retinal organoids. *Developmental Cell* 57, 820-836.e6 (2022).
8. Li, K., Wei, G.-H., Yin, Y. & Feng, J. Targeting Noncoding cis-Regulatory Elements for Cancer Therapy in the Context of the 3D Genome. *Cancer Discov* 14, 2061–2065 (2024).
9. Schmitz, R. J., Grotewold, E. & Stam, M. Cis-regulatory sequences in plants: Their importance, discovery, and future challenges. *Plant Cell* 34, 718–741 (2022).
10. Wang, H. *et al.* De novo design of insulated cis-regulatory elements based on

- deep learning-predicted fitness landscape. *Nucleic Acids Res* 53, gkaf611 (2025).
11. de Almeida, B. P., Reiter, F., Pagani, M. & Stark, A. DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. *Nat Genet* 54, 613–624 (2022).
 12. Taskiran, I. I. *et al.* Cell-type-directed design of synthetic enhancers. *Nature* 626, 212–220 (2024).
 13. Zhang, P. *et al.* Deep flanking sequence engineering for efficient promoter design using DeepSEED. *Nat. Commun.* 14, 6309 (2023).
 14. Xia, Y., Du, X., Liu, B., Guo, S. & Huo, Y.-X. Species-specific design of artificial promoters by transfer-learning based generative deep-learning model. *Nucleic Acids Res.* 52, 6145–6157 (2024).
 15. Lawler, A. J. *et al.* Machine learning sequence prioritization for cell type-specific enhancer design. *eLife* 11, e69571 (2022).
 16. Xia, Y., Du, X., Liu, B., Guo, S. & Huo, Y.-X. Species-specific design of artificial promoters by transfer-learning based generative deep-learning model. *Nucleic Acids Research* 52, 6145 (2024).
 17. Li, T. *et al.* Modeling 0.6 million genes for the rational design of functional cis-regulatory variants and de novo design of cis-regulatory sequences. *Proceedings of the National Academy of Sciences* 121, e2319811121 (2024).
 18. Li, Z. *et al.* A novel interpretable deep learning-based computational framework designed synthetic enhancers with broad cross-species activity. *Nucleic Acids Research* gkae912 (2024) doi:10.1093/nar/gkae912.

19. Gosai, S. J. *et al.* Machine-guided design of cell-type-targeting cis-regulatory elements. *Nature* 634, 1211–1220 (2024).
20. de Almeida, B. P. *et al.* Targeted design of synthetic enhancers for selected tissues in the *Drosophila* embryo. *Nature* 626, 207–211 (2024).
21. Fu, X. *et al.* A foundation model of transcription across human cell types. *Nature* 637, 965–973 (2025).
22. Lal, A., Garfield, D., Biancalani, T. & Eraslan, G. Designing realistic regulatory DNA with autoregressive language models. *Genome Res.* <https://doi.org/10.1101/gr.279142.124> (2024) doi:10.1101/gr.279142.124.
23. Jaganathan, K. *et al.* Predicting expression-altering promoter mutations with deep learning. *Science* 0, eads7373 (2025).
24. Qiu, Y. *et al.* Precise engineering of gene expression by editing plasticity. *Genome Biology* 26, 51 (2025).
25. Zhou, Z. *et al.* DNABERT-2: Efficient Foundation Model and Benchmark For Multi-Species Genome. Preprint at <https://doi.org/10.48550/arXiv.2306.15006> (2024).
26. Sanabria, M., Hirsch, J., Joubert, P. M. & Poetsch, A. R. DNA language model GROVER learns sequence context in the human genome. *Nat Mach Intell* 6, 911–923 (2024).
27. Liu, G. *et al.* PDLLMs: A group of tailored DNA large language models for analyzing plant genomes. *Molecular Plant* 18, 175–178 (2025).
28. Mendoza-Revilla, J. *et al.* A foundational large language model for edible plant

- genomes. *Commun Biol* 7, 1–18 (2024).
29. Dalla-Torre, H. *et al.* Nucleotide Transformer: building and evaluating robust foundation models for human genomics. *Nat Methods* 1–11 (2024) doi:10.1038/s41592-024-02523-z.
30. Nguyen, E. *et al.* Sequence modeling and design from molecular to genome scale with Evo. *Science* 386, eado9336 (2024).
31. Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Preprint at <https://doi.org/10.48550/arXiv.1810.04805> (2019).
32. Lin, C.-C., Jaech, A., Li, X., Gormley, M. R. & Eisner, J. Limitations of Autoregressive Models and Their Alternatives. Preprint at <https://doi.org/10.48550/arXiv.2010.11939> (2021).
33. Buenrostro, J. D., Giresi, P. G., Zaba, L. C., Chang, H. Y. & Greenleaf, W. J. Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position. *Nat Methods* 10, 1213–1218 (2013).
34. Boyle, A. P. *et al.* High-Resolution Mapping and Characterization of Open Chromatin across the Genome. *Cell* 132, 311–322 (2008).
35. Black, S., Leo, G., Wang, P., Leahy, C. & Biderman, S. GPT-Neo: Large Scale Autoregressive Language Modeling with Mesh-Tensorflow. Zenodo <https://doi.org/10.5281/zenodo.5297715> (2021).
36. Ricci, W. A. *et al.* Widespread long-range cis-regulatory elements in the maize

- genome. *Nat. Plants* 5, 1237–1249 (2019).
37. Zhao, H. *et al.* Explainability for Large Language Models: A Survey. Preprint at <https://doi.org/10.48550/arXiv.2309.01029> (2023).
38. Jores, T. *et al.* Synthetic promoter designs enabled by a comprehensive analysis of plant core promoters. *Nat. Plants* 7, 842–855 (2021).
39. Du, F. *et al.* A Survey of LLM Datasets: From Autoregressive Model to AI Chatbot. *J. Comput. Sci. Technol.* 39, 542–566 (2024).
40. Sennrich, R., Haddow, B. & Birch, A. Neural Machine Translation of Rare Words with Subword Units. in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (eds. Erk, K. & Smith, N. A.) 1715–1725 (Association for Computational Linguistics, Berlin, Germany, 2016). doi:10.18653/v1/P16-1162.
41. Ding, K. *et al.* PlantCADB: A Comprehensive Plant Chromatin Accessibility Database. *Genomics, Proteomics & Bioinformatics* 21, 311–323 (2023).
42. Kingma, D. P. & Ba, J. Adam: A Method for Stochastic Optimization. Preprint at <https://doi.org/10.48550/arXiv.1412.6980> (2017).
43. Hu, E. J. *et al.* LoRA: Low-Rank Adaptation of Large Language Models. Preprint at <https://doi.org/10.48550/arXiv.2106.09685> (2021).
44. Grant, C. E., Bailey, T. L. & Noble, W. S. FIMO: scanning for occurrences of a given motif. *Bioinformatics* 27, 1017–1018 (2011).
45. Bailey, T. L. & Elkan, C. Fitting a mixture model by expectation maximization to discover motifs in biopolymers. *Proc Int Conf Intell Syst Mol Biol* 2, 28–36 (1994).

46. Gupta, S., Stamatoyannopoulos, J. A., Bailey, T. L. & Noble, W. S. Quantifying similarity between motifs. *Genome Biology* 8, R24 (2007).